

Assignment: Remote Desktop Services (RDS) Configuration in Windows Server

Task 1: Install the Remote Desktop Services (RDS) Role

Steps:

1. Open **Server Manager** on the Windows Server VM.
2. Click **Manage → Add Roles and Features**.
3. Click **Next** until the **Installation Type** page appears.
4. Select **Role-based or feature-based installation**.
5. Click **Next**.
6. Select the local Windows Server from the **Server Pool**.
7. Click **Next** until the **Server Roles** page appears.
8. Select **Remote Desktop Services**.
9. When prompted, click **Add Features**.
10. Click **Next** through the remaining pages.
11. On the **Confirmation** page, click **Install**.
12. Wait for the installation to complete.
13. Click **Close**.

Result:

The **Remote Desktop Services (RDS)** role is successfully installed on the Windows Server.

Task 2: Create an RDS Session Collection and Publish RemoteApps

Steps:

1. Open **Server Manager**.
2. Click **Remote Desktop Services** from the left-side menu.
3. Open the **Overview** page.
4. Configure the required RDS deployment components, including:
 - RD Connection Broker

- RD Web Access
 - RD Session Host
5. After the RDS deployment is configured, open **Collections**.
 6. Click **Tasks → Create Session Collection**.
 7. Enter the Collection Name as:

StudentApps

8. Click **Next**.
9. Select the appropriate **RD Session Host server**.
10. Click **Next**.
11. Specify the users or groups who are allowed to access the collection.
12. Click **Next**.
13. Review the configuration.
14. Click **Create**.
15. Wait for the collection creation to complete.

Publish Notepad and Calculator as RemoteApps

1. Open **Remote Desktop Services → Collections**.
2. Select the **StudentApps** collection.
3. Click **Tasks → Publish RemoteApps**.
4. Select **Notepad** from the available applications.
5. Select **Calculator**.
6. Click **Next**.
7. Review the selected applications.
8. Click **Publish**.
9. Wait for the publishing process to complete.
10. Verify that **Notepad** and **Calculator** appear under the published RemoteApps.

Result:

An RDS Session Collection named **StudentApps** is successfully created, and **Notepad** and **Calculator** are published as RemoteApps for authorized users.

Task 3: Create a Local User and Verify Remote Desktop Access

Part A: Create the "testuser" Account

Steps:

1. Open **Server Manager**.
2. Go to **Tools → Computer Management**.
3. Expand **Local Users and Groups**.
4. Select **Users**.
5. Right-click **Users** and select **New User**.
6. Enter the following details:

User Name: testuser

Password: Enter a suitable password

7. Configure the required password options.
8. Click **Create**.
9. Click **Close**.

Result:

A local user account named **testuser** is successfully created.

Part B: Add testuser to the Remote Desktop Users Group

Steps:

1. In **Computer Management**, expand **Local Users and Groups**.
2. Select **Groups**.
3. Double-click **Remote Desktop Users**.
4. Click **Add**.
5. Enter:

testuser

6. Click **Check Names**.
7. Click **OK**.
8. Click **Apply** and then **OK**.

Result:

The **testuser** account is successfully added to the **Remote Desktop Users** group.

Part C: Enable Remote Desktop

Steps:

1. Open **Server Manager**.
2. Open **Settings → System → Remote Desktop**.
3. Turn on **Remote Desktop**.
4. Confirm the setting when prompted.
5. Make sure **Network Level Authentication (NLA)** is enabled if required.
6. Note the server's IP address by opening Command Prompt and running:

ipconfig

7. Note the **IPv4 Address** of the server.
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Part D: Connect from Another Computer

Steps:

1. On another Windows computer connected to the same network, press:

Windows Key + R

2. Type:

mstsc

3. Press **Enter**.
4. In the Remote Desktop Connection window, enter the Windows Server's IP address.

5. Click **Connect**.
6. Enter the credentials of:

Username: testuser

Password: The password created for testuser

7. Accept the certificate warning if it appears.
8. Wait for the Remote Desktop session to connect.

Result:

The **testuser** account successfully connects to the Windows Server through Remote Desktop.

Task 4: Simulate Multiple Remote Desktop Sessions

Objective:

To verify how the RDS server handles multiple users connecting at the same time.

Steps:

1. Make sure the RDS server is running.
2. From another computer, open **Remote Desktop Connection** using:

mstsc

3. Connect to the server using the **Administrator** account.
4. Open another Remote Desktop Connection window.
5. Connect to the same server using the **testuser** account.
6. Keep both Remote Desktop sessions active.
7. On the server, open **Task Manager**.
8. Click the **Users** tab.
9. Observe the currently logged-in users and their sessions.
10. Check the **CPU**, **Memory**, and other resource usage.
11. Open applications such as **Notepad** or **Calculator** in both sessions.
12. Observe whether both sessions continue to work simultaneously.

Observation:

User	Session Status Observation	
Administrator	Connected	Administrator session is active.
testuser	Connected	testuser session is active simultaneously.

Resource Observation:

When multiple users are connected at the same time, the server consumes additional **CPU and RAM resources**. The actual performance depends on the resources allocated to the Windows Server VM.

Session Limitation:

If the server is configured only for normal Windows Server Remote Desktop administration, there may be limitations on the number of simultaneous administrative sessions. A properly configured **Remote Desktop Services deployment with RD Session Host** is used when multiple users need regular Remote Desktop sessions.

Result:

Multiple Remote Desktop sessions were successfully tested using different user accounts. The server handled simultaneous connections, while CPU and memory usage increased according to the number of active sessions and applications.